Bishal Shrestha

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Research Interests

Artificial intelligence and computational biology, deep learning models for genomics and 3D chromatin organization, single-cell genomics, graph neural networks and transformers for biological data, protein structure prediction and function annotation, spatiotemporal analysis of Hi-C data, and developing computational tools for understanding genome architecture and its role in human health and disease.

Research Experience

Research Assistant Aug 2023 – Present
University of Miami, Coral Gables, FL

- Developed HiC4D-SPOT, an unsupervised deep-learning framework using ConvLSTM-based autoencoders to detect spatiotemporal anomalies in Hi-C data, published in *Briefings in Bioinformatics* (2025); achieved high reconstruction fidelity (PCC/SCC > 0.9) and accurately identified structural anomalies including TAD disruptions and chromatin remodeling events.
- Developed scHiGex, a Graph Transformer-based model for predicting single-cell gene expression using simultaneously captured single-cell Hi-C data, published in *NAR Genomics and Bioinformatics* (2025); achieved average absolute error of 0.07 and precise cell type classification (adjusted Rand index = 1).
- Designed and implemented an ensemble of graph neural network architectures to estimate local residue accuracy in protein structure models as part of the CASP 16, a leading competition in computational biology.
- Implemented Geometric Algebra Transformers to annotate protein functions for protein models, showcasing use of geometric deep learning in structural biology.

Research Intern Sep 2021 – Aug 2022
University of Missouri - St. Louis, MO

- Conducted research on adversarial sample generation for Nepal's embossed license plate recognition system, employing adversarial training to enhance model robustness against attack scenarios.
- Designed and executed deep learning experiments using convolutional neural networks (CNNs) in PyTorch, including collecting and preprocessing a novel dataset for real-world applications.

Student Researcher Apr 2021 – Sep 2021
Research and Innovation Unit, Tribhuvan University

- Investigated machine learning techniques for customer churn prediction, collaborating with a leading Nepali ISP to optimize business operations through predictive analytics.
- Presented findings at the Himalaya International Conference, highlighting the impact of academia-industry partnerships in addressing real-world problems.

Education

PhD in Computer Science Aug 2023 - Present
University of Miami, Coral Gables, FL *Advisor: Dr. Zheng Wang*

Bachelor's in Computer Engineering Jun 2017 - Jan 2022
Tribhuvan University, Nepal *Advisor: Dr. Badri Adhikari*

Teaching

Visiting Lecturer Spring 2025, Fall 2024
University of Miami, Coral Gables, FL

- **CSC314: Computer Organization and Architecture**: Spring 2025, Fall 2024 (2 sections)

Graduate Teaching Assistant Spring 2025, Spring 2024, Fall 2023
University of Miami, Coral Gables, FL

- **CSC402: Computer Science Practicum II**: Spring 2025, Spring 2024
  Student Course Evaluations: Achieved **4.9/5** (Spring 2024) and **4.8/5** (Spring 2025)
- **CSC113/CSC200: Data Science for the World**: Fall 2023

Programming Skills

Programming Languages: Python, C, C++

Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

Data Visualization: Matplotlib, Seaborn

Certifications: Mathematics for Machine Learning (Imperial College London), Neural Networks and Deep Learning (Coursera), Qiskit Global Summer School (IBM)

Soft Skills: Effective communication, teamwork, problem-solving, critical thinking

References

- **Dr. Zheng Wang**: Associate Professor, Department of Computer Science, University of Miami
  Email: zheng.wang@miami.edu; Phone: +1 (305) 284-3642
  Address: 330M Ungar Building, 1365 Memorial Drive, Coral Gables, FL 33124-4245
  (PhD Advisor)
- **Dr. Badri Adhikari**: Associate Professor, Department of Computer Science, University of Missouri - St. Louis
  Email: adhikarib@umsl.edu; Phone: +1 (573) 639-2847
  Address: Department of Computer Science, University of Missouri - St. Louis, 1 University Boulevard, St. Louis, MO 63121
  (Research Mentor)

Publications

- **Shrestha B**, Wang Z. *"HiC4D-SPOT: a spatiotemporal outlier detection tool for Hi-C data."* Briefings in Bioinformatics, 26(4), bbaf341. 2025.
- **Shrestha B**, Siciliano AJ, Zhu H, Liu T, Wang Z. *"scHiGex: predicting single-cell gene expression based on single-cell Hi-C data."* NAR Genomics and Bioinformatics. 7 (1), Iqaf002. 2025.
- **Shrestha B**, Khakurel G, Simkhada K, Adhikari B. *"Adversarial sample generation and training using geometric masks for accurate and resilient license plate character recognition."* arXiv preprint arXiv:2311.12857. 2023.
- Prajapati S, Engelhardt M, Shrestha A, Khakurel G, **Shrestha B**, Simkhada K. *"Industry-Academia Collaboration in Nepal - A case study of Customer Churn Prediction for Worldlink."* Research and Innovation Unit. 2021.

Publications Under Review

10. Di Giovanni S, Mueller F, Maclachlan E, Costa A, Qu J, **Shrestha B**, Wang Z, De Virgiliis F, Hutson T, Zhou L, Kong G, Chadwick J, Sousa M, Palmisano I. *"Druggable CITED2 Is a Reversible Epigenetic Switch in Neuronal Maturation to Regenerative Decline."* EMBO Molecular Medicine, Resubmitted Dec 2025.
11. Siciliano AJ, Bao Y, **Shrestha B**, Wang Z. *"Inferring the qualities of protein-RNA models with graph transformers."* Bioinformatics, Submitted Nov 2025.

Poster and Presentations

1. **Shrestha B** (Advisor: Z. Wang) *"HiC4D-SPOT: a spatiotemporal outlier detection tool for Hi-C data"* Research Presentation at the annual Computing Day, University of Miami, Coral Gables, FL, April 2025.
2. **Shrestha B** (Advisor: Z. Wang) *"Predicting single-cell gene expression based on single-cell Hi-C using graph transformer"* Research Presentation at the annual Computing Day, University of Miami, Coral Gables, FL, April 2024.
3. **Shrestha B** (Advisor: Z. Wang) *"Predicting single-cell gene expression based on single-cell Hi-C using graph transformer"* Research Presentation at the Sylvester Annual Retreat, Sylvester Comprehensive Cancer Center, University of Miami, Miami, FL, Oct 2024.